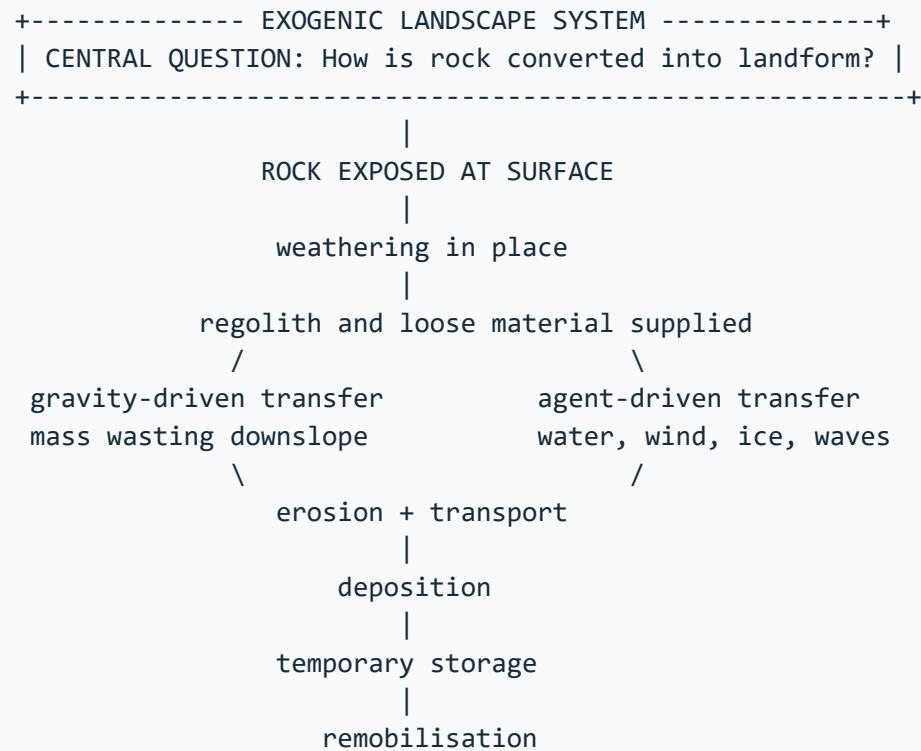


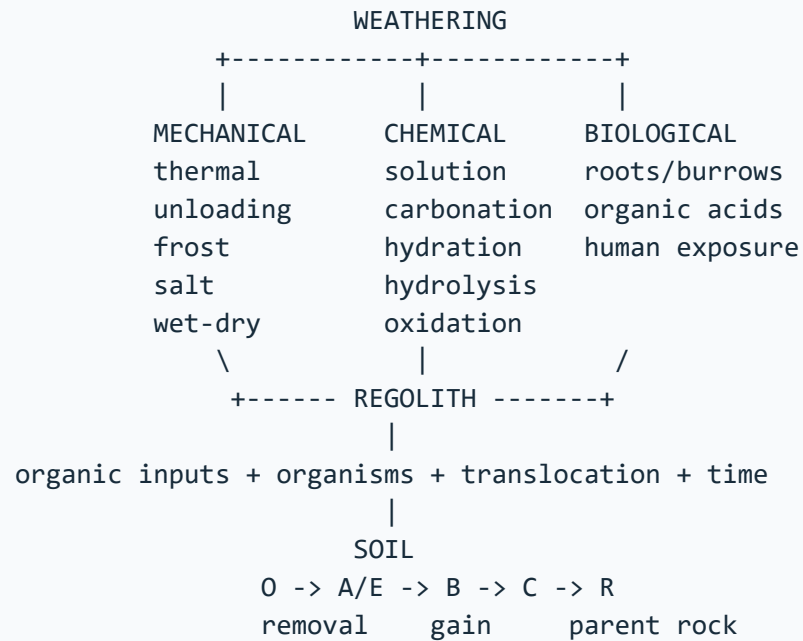
ASCII MASTER FLOW – PANEL 1/12: Exogenic source-transfer-storage system



Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

MUST REMEMBER: Weathering prepares material in situ, mass movement transfers it downslope, running water removes it, and groundwater follows infiltration, percolation, aquifer storage and discharge; Indian analysis links lithology, slope, rainfall, land use and pumping.

ASCII MASTER FLOW – PANEL 2/12: Weathering taxonomy, controls, regolith and soil formation



CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief
+ drainage + organisms + time + land use

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks;
parent material does not alone determine every soil property.

ASCII MASTER FLOW – PANEL 3/12: Soil erosion sequence and watershed control logic

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION

```

      |
diffuse sheet removal
      |
small rill channels
      |
entrenched gullies
      |
ravine landscape: Chambal cue
```

WATERSHED RESPONSE

cover soil -> slow runoff -> shorten slope length -> contour/terrace
-> protect drainage -> treat gully head -> rehabilitate catchment

WIND BRANCH

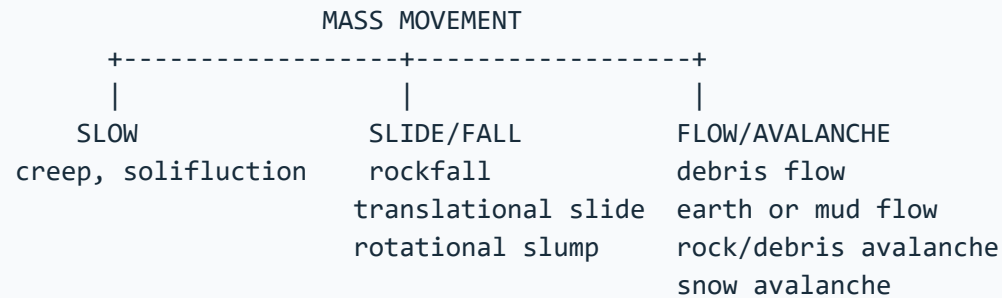
surface creep -> saltation -> suspension

COASTAL BRANCH

wave erosion belongs to the coastal-process system; use only when asked.

A check dam alone cannot replace upstream cover, drainage and maintenance.

ASCII MASTER FLOW – PANEL 4/12: Mass-movement taxonomy and slope-stability mechanism



SLOPE BALANCE

resisting strength = cohesion + friction + structural support

versus

driving stress = gravity component + loading + toe loss

rain infiltration -> added weight + pore pressure -> lower effective strength

road or river cut -> steepening / undercutting -> support loss

quake / quarry / leakage / drainage change -> trigger or amplification

FAILURE WHEN DRIVE > RESISTANCE

Material, movement and water define the event; speed alone does not.

ASCII MASTER FLOW – PANEL 5/12: India landslide geography, risk production and mitigation stack

INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity
Western Ghats : escarpment + deep weathering + orographic rain + cuts
Nilgiris : weathered slopes + roads + settlements + drainage change

|
PREDISPOSITION -> TRIGGER -> MOVEMENT -> EXPOSURE -> LOSS
geology/slope rain/quake slide/flow road/town disaster

GSI LADDER

|
inventory -> susceptibility/hazard study -> post-event diagnosis
-> monitoring / NLFC support -> decision communication

MITIGATION STACK

|
avoid unsafe siting -> manage drainage -> protect toe
-> stabilise slope -> bioengineer -> control spoil/quarrying
-> maintain works -> warn/close routes -> evacuate and prepare

Inventory maps past events; susceptibility maps relative likelihood;
neither is a deterministic event prediction.
Joshimath is a multi-causal subsidence case, not a synonym for landslide.

ASCII MASTER FLOW – PANEL 6/12: Groundwater architecture from recharge to wells

rain -> infiltration -> soil moisture -> percolation -> recharge

```
GROUND SURFACE      recharge area              spring at slope
-----v----->
ZONE OF AERATION    perched water on local aquitard
----- capillary fringe -----
WATER TABLE        UNCONFINED AQUIFER          pumping well
~~~~~\_____ cone of depression
AQUITARD             slow transmission
-----
CONFINED AQUIFER     pressure; water rises to potentiometric level
=====
AQUICLUDE            practically no transmission
```

porosity = fraction of voids | permeability = connected flow capacity
aquifer stores and transmits | aquitard transmits slowly
artesian water rises under pressure; it flows only if head exceeds ground

Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

ASCII MASTER FLOW – PANEL 7/12: India groundwater depletion, quality and subsidence geography

REGION	AQUIFER FORM	MAJOR PRESSURE
Indo-Gangetic	layered alluvium	depletion, As, nitrate
Peninsular	weathered fractures	discontinuous storage
Deccan basalt	contacts + fractures	uneven well yield
Arid hard rock	limited recharge	fluoride, salinity
Coastal belt	fresh-saline wedge	intrusion and upconing

crop-water demand + cheap energy + private wells + urban growth
+ weak aquifer regulation + limited recharge

falling head -> lower yield -> higher lift cost -> unequal access

compressible sediment: pore-pressure loss -> compaction -> subsidence

Quality paths differ: arsenic redox | fluoride rock-water interaction

nitrate sewage/fertiliser | salinity coastal or inland

Quantity and quality require separate evidence, though each limits usable water.

ASCII MASTER FLOW – PANEL 8/12: Recharge, demand governance and integrated Mains answer spine

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

SUPPLY / RECHARGE

roof and runoff capture

managed aquifer recharge

watershed treatment

spring-shed restoration

quality-safe source water

maintenance and monitoring

DEMAND / GOVERNANCE

crop-water alignment

efficient irrigation

well registration/metering

conjunctive use and reuse

community water budgets

enforceable state/local rules

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/
+---- AQUIFER PLAN ----+

```

CGWB assessment + NAQUIM mapping + state regulation

+ Atal Bhujal learning + local institutions + public data

ANSWER SPINE

1 define stock-flow-quality system | 2 draw aquifer or India map

3 diagnose regional mechanism | 4 use dated evidence carefully

5 match intervention to cause | 6 qualify limits and equity

Recharge structures are inputs, not proof of water-table recovery.

Desalination supplies water but does not restore freshwater aquifer head.

ASCII MASTER FLOW – PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.

ASCII MASTER FLOW – PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Causal method Initial condition -> energy/force or driver ->...
2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT
3. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,...

VERDICT: Qualification earns marks; false certainty is a hard failure.

CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface, aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

ASCII MASTER FLOW – PANEL 11/12: Integrated answer spine and qualified conclusion

ANSWER SPINE

1. Original solved Mains questions 6
2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023,...
3. Practice contract Every solved item has demand decoding, a detailed...

VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

EVIDENCE LIMIT: A trigger such as intense rain is not the complete cause: antecedent moisture, discontinuities, toe cutting, drainage and construction condition failure; groundwater trends and notified areas require dated official status.

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.